

Tables

Reported biomarker–cognition associations do not translate into predictive utility: a pre-registered analysis of NHANES 2011–2014

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Table 1: Participant characteristics, analytic sample ($n = 1,784$), overall and by outcome status.

Characteristic	Overall	Low cognitive performance	Not low cognitive performance
<i>n</i>	1784	431	1353
Age, years, mean (SD)	69.37 (6.74)	72.07 (6.94)	68.52 (6.44)
Income-to-poverty ratio, mean (SD)	2.60 (1.60)	1.91 (1.38)	2.82 (1.61)
Blood lead, $\mu\text{g/dL}$, median [IQR]	1.47 [1.01, 2.19]	1.57 [1.03, 2.41]	1.44 [1.01, 2.12]
Blood cadmium, $\mu\text{g/L}$, median [IQR]	0.38 [0.25, 0.61]	0.42 [0.27, 0.69]	0.37 [0.25, 0.58]
Blood total mercury, $\mu\text{g/L}$, median [IQR]	0.93 [0.49, 1.94]	0.75 [0.40, 1.56]	0.99 [0.53, 2.01]
Blood selenium, $\mu\text{g/L}$, median [IQR]	193.09 [177.40, 208.38]	188.35 [173.17, 204.15]	194.23 [179.15, 209.06]
Blood manganese, $\mu\text{g/L}$, median [IQR]	8.71 [6.90, 11.09]	8.35 [6.50, 10.69]	8.80 [7.02, 11.19]
Serum vitamin B12, pmol/L , median [IQR]	410.30 [289.30, 601.45]	421.40 [284.50, 621.80]	408.10 [290.00, 599.30]
Serum methylmalonic acid, $\mu\text{mol/L}$, median [IQR]	163.00 [125.75, 227.00]	181.00 [131.00, 256.00]	160.00 [124.00, 214.00]
Serum total folate, nmol/L , median [IQR]	20.80 [13.60, 31.42]	20.00 [13.05, 32.05]	20.90 [13.70, 31.30]
RBC folate, nmol/L , median [IQR]	543.00 [398.00, 737.00]	530.00 [391.00, 748.50]	547.00 [401.00, 733.00]
25-hydroxyvitamin D, nmol/L , mean (SD)	76.31 (30.44)	72.68 (30.36)	77.46 (30.39)
HbA1c, %, median [IQR]	5.80 [5.50, 6.30]	5.90 [5.50, 6.50]	5.80 [5.50, 6.20]
HDL cholesterol, mg/dL , median [IQR]	52.00 [43.00, 63.00]	49.00 [40.00, 60.00]	52.00 [44.00, 63.00]
Total cholesterol, mg/dL , mean (SD)	192.85 (43.15)	185.58 (41.41)	195.16 (43.45)
Triglycerides, mg/dL , median [IQR]	126.00 [86.00, 187.25]	128.00 [91.00, 182.00]	126.00 [85.00, 189.00]
Serum albumin, g/dL , mean (SD)	4.20 (0.31)	4.15 (0.34)	4.21 (0.29)
Serum iron, $\mu\text{g/dL}$, median [IQR]	80.00 [61.00, 103.00]	76.00 [57.00, 96.00]	82.00 [63.00, 104.00]
eGFR, mL/min/1.73 m^2 , mean (SD)	75.64 (19.85)	69.95 (22.19)	77.45 (18.69)
Neutrophil-to-lymphocyte ratio, median [IQR]	2.11 [1.55, 2.92]	2.35 [1.67, 3.21]	2.05 [1.53, 2.83]
Platelet-to-lymphocyte ratio, median [IQR]	119.80 [94.11, 152.96]	119.05 [89.76, 153.87]	120.00 [95.26, 152.50]
Systemic immune-inflammation index, median [IQR]	458.83 [319.14, 664.17]	480.65 [322.90, 714.20]	452.00 [315.00, 646.00]
Body mass index, kg/m^2 , median [IQR]	28.10 [24.77, 32.40]	27.60 [24.30, 31.90]	28.20 [24.90, 32.50]
Alcoholic drinks per day, median [IQR]	0.00 [0.00, 1.25]	0.00 [0.00, 1.00]	1.00 [0.00, 2.00]
PHQ-9 depression score, median [IQR]	1.00 [0.00, 4.00]	2.00 [0.00, 6.50]	1.00 [0.00, 4.00]
Sex = Male	880 (49.3%)	238 (55.2%)	642 (47.5%)
Sex = Female	904 (50.7%)	193 (44.8%)	711 (52.5%)
Race/ethnicity = Mexican American	140 (7.8%)	46 (10.7%)	94 (6.9%)
Race/ethnicity = Other Hispanic	182 (10.2%)	68 (15.8%)	114 (8.4%)

Table 1 continued

Characteristic	Overall	Low cognitive performance	Not low cognitive performance
Race/ethnicity = Non-Hispanic White	863 (48.4%)	165 (38.3%)	698 (51.6%)
Race/ethnicity = Non-Hispanic Black	425 (23.8%)	120 (27.8%)	305 (22.5%)
Race/ethnicity = Non-Hispanic Asian	144 (8.1%)	26 (6.0%)	118 (8.7%)
Race/ethnicity = Other/multiracial	30 (1.7%)	6 (1.4%)	24 (1.8%)
Education = Less than 9th grade	224 (12.6%)	137 (31.8%)	87 (6.4%)
Education = 9–11th grade	219 (12.3%)	79 (18.3%)	140 (10.3%)
Education = High school grad/GED	417 (23.4%)	103 (23.9%)	314 (23.2%)
Education = Some college/AA degree	511 (28.6%)	71 (16.5%)	440 (32.5%)
Education = College graduate or above	413 (23.2%)	41 (9.5%)	372 (27.5%)
Smoking status = Never	895 (50.2%)	210 (48.7%)	685 (50.6%)
Smoking status = Former	671 (37.6%)	159 (36.9%)	512 (37.8%)
Smoking status = Current	218 (12.2%)	62 (14.4%)	156 (11.5%)
Ever told had a stroke	125 (7.0%)	49 (11.4%)	76 (5.6%)
Ever told had diabetes	490 (27.5%)	154 (35.7%)	336 (24.8%)
Ever told had high blood pressure	1111 (62.3%)	302 (70.1%)	809 (59.8%)
Ever told had high cholesterol	997 (55.9%)	234 (54.3%)	763 (56.4%)

Table 2: Included versus excluded participants, among the 3,124 outcome-eligible participants.

Characteristic	Included (<i>n</i> = 1,784)	Excluded (<i>n</i> = 1,340)
Age, years, mean (SD)	69.37 (6.74)	70.10 (6.94)
Income-to-poverty ratio, mean (SD)	2.60 (1.60)	2.50 (1.59)
PHQ-9 depression score, median [IQR]	1.00 [0.00, 4.00]	2.00 [0.00, 5.00]
Female, %	50.7%	52.6%
Low cognitive performance, %	24.2%	26.1%
Diabetes, %	27.5%	30.0%